

10T3: MTG-UPF

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Music Technology Group - UPF

- part of the Department of Information and Communication Technologies
- Teaching:
 - Master in Sound and Music Computing
 - PhD in Information and Communication Technologies
- Research:
 - Audio Signal Processing
 - Sound and Music Description
 - Musical and Advanced Interaction
 - Semantic Technologies for Sound and Music

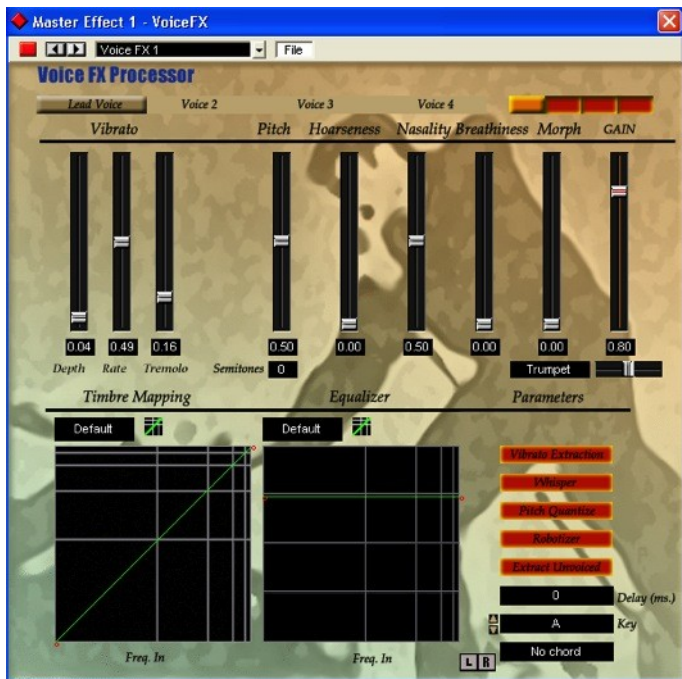
Audio signal processing

Singing voice synthesis



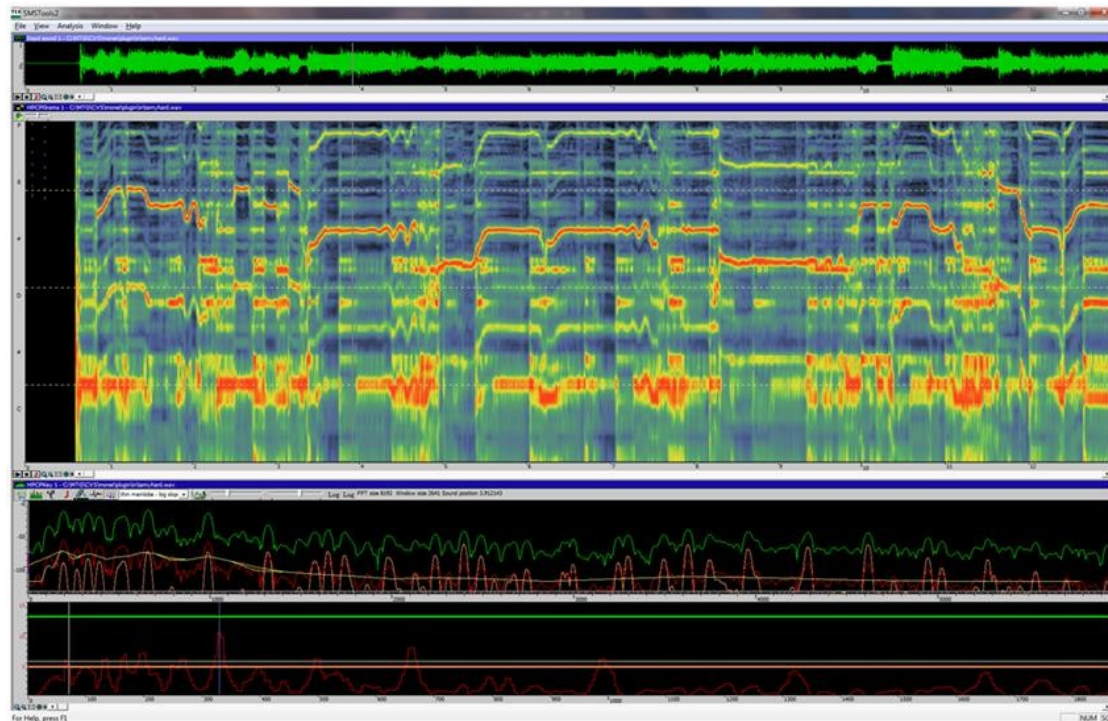
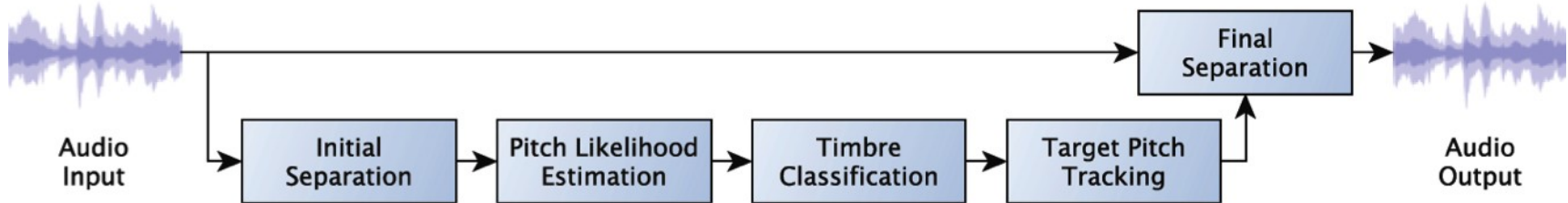
Audio signal processing

Voice transformations



Audio signal processing

Sound source separation



Sound and music description

Audio features

```
metadata:  
  version:  
    essentia: "1.2.2"
```

```
pitch_confidence:  
  mean: 0.725613296032  
  skew: -1.01243495941
```

```
spectral_flatness_db:  
  mean: 0.189157426357
```

```
spectral_rms:  
  mean: 0.0014992490178
```

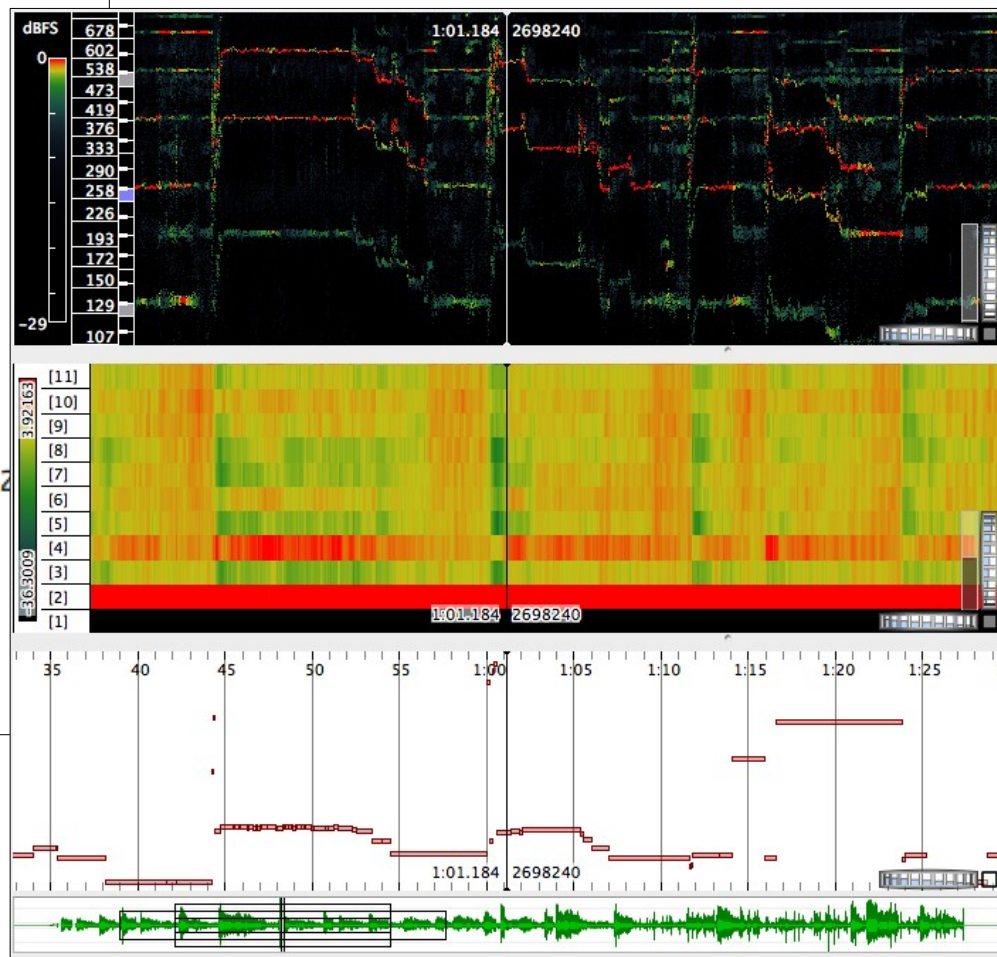
```
spectral_strongpeak:  
  mean: 0.628910064697  
  var: 0.741435945034
```

```
mfcc:  
  mean: [143.017318726, -38.4358482361, 7.88732626]  
  var: [122.917526245, 159.187026978]
```

```
onset_rate: [1.70000004768]
```

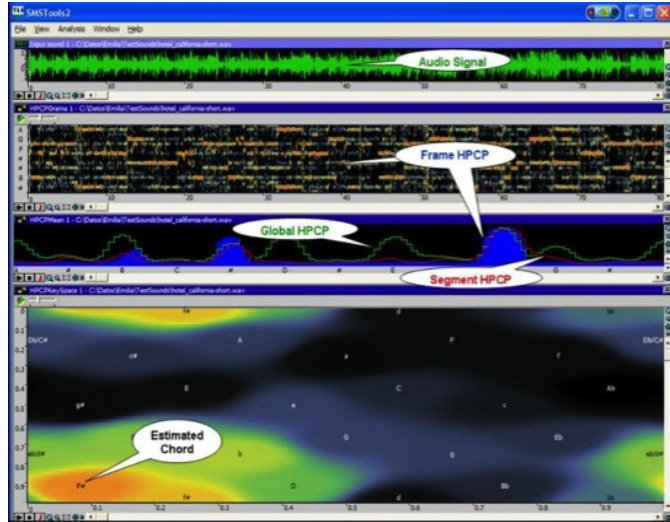
```
tristimulus:  
  mean: [0.29845097661, 0.602976679802]
```

 SSENTIA

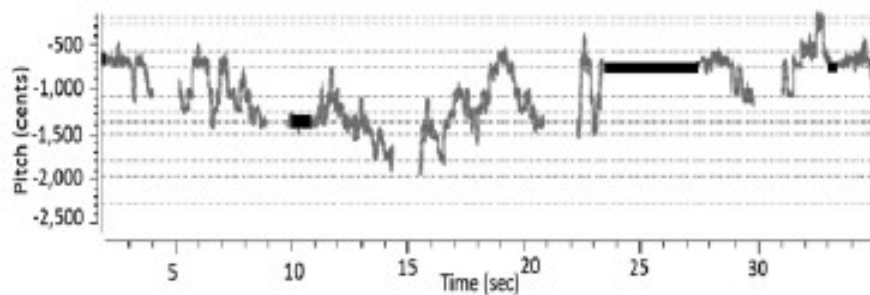


Sound and music description

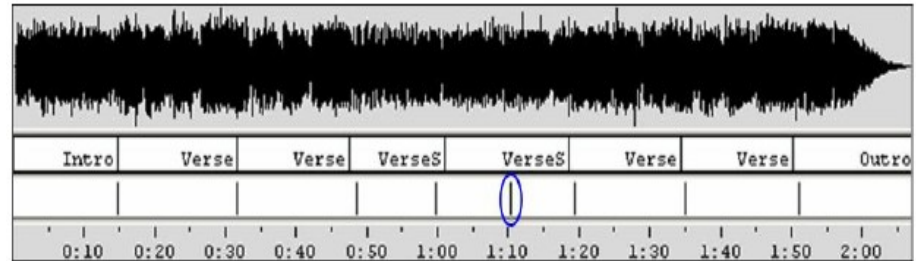
Musical features



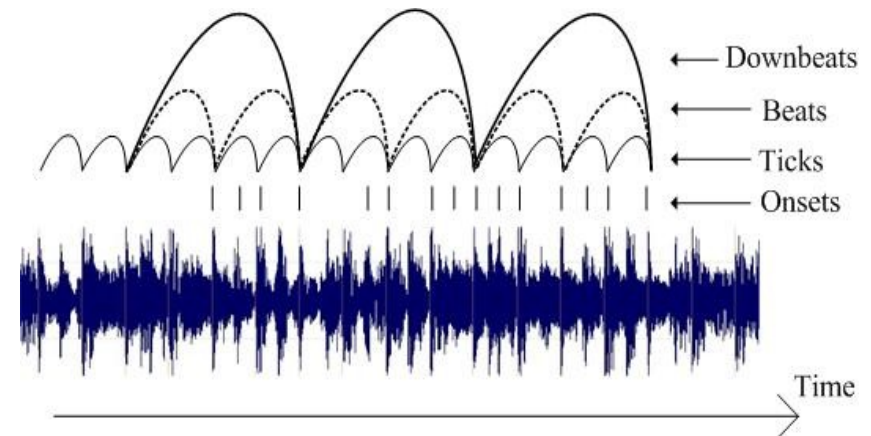
harmony



melody



structure



rhythm

Sound and music description

Music collections

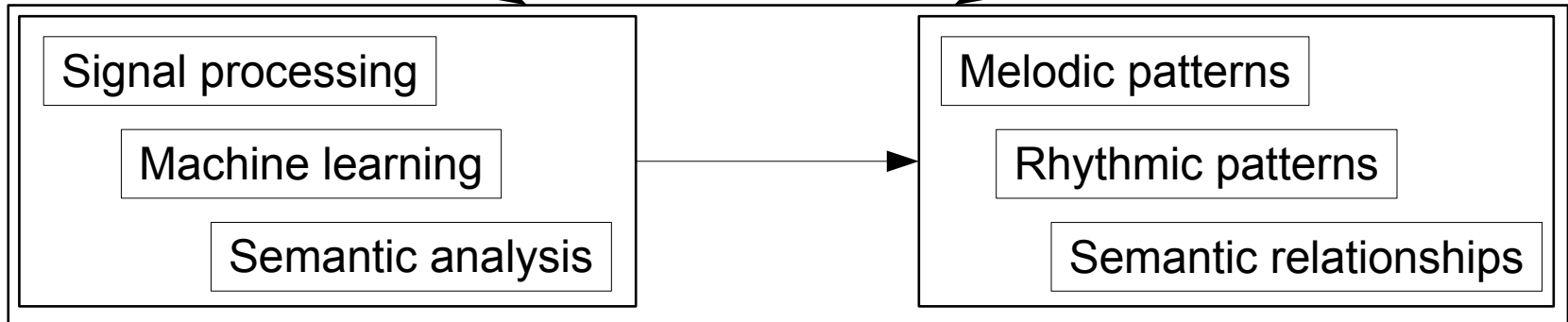


Dunya Hindustani
— Public collection by [?] compmusic (See all of compmusic's public collections)

Overview

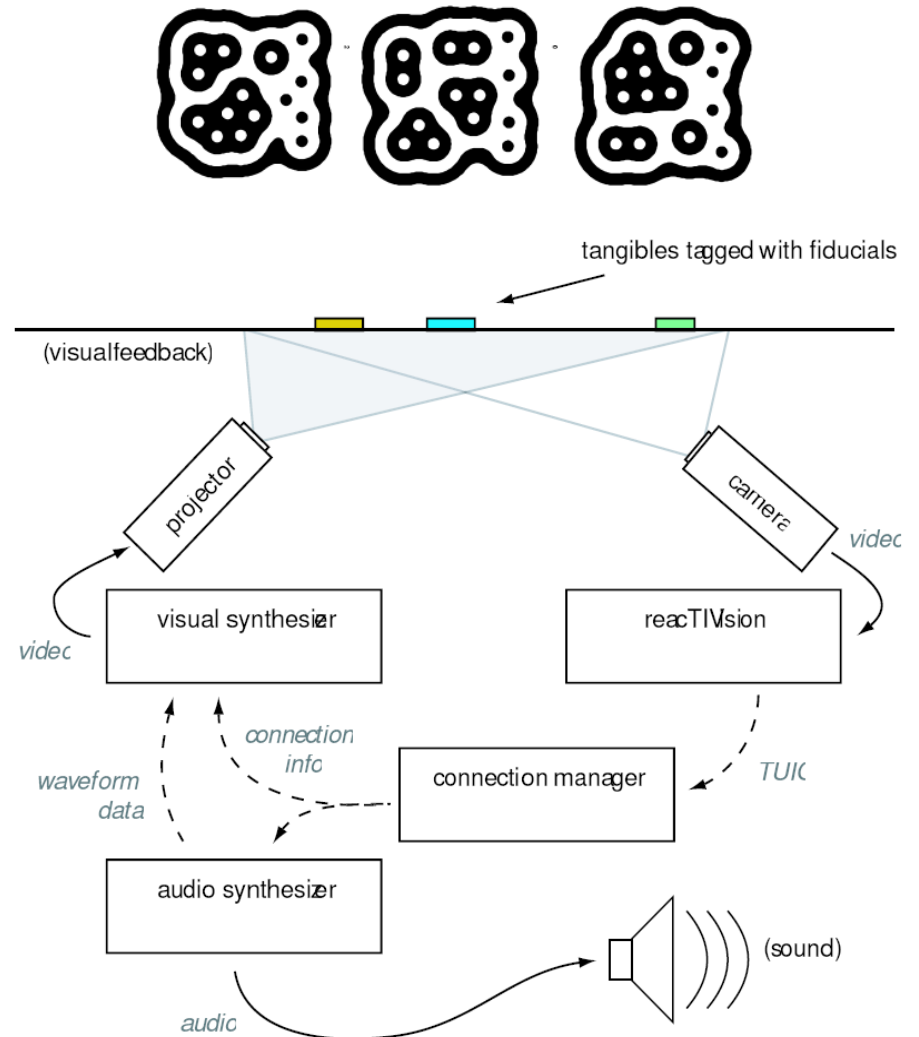
Releases

Release	Artist	Format	Tracks	Date	Country	Label	Catalog#
All India Radio Archival Release Vol. 3	Vijayal Hussain Khan	CD	2	1960	IN	T-Series	SVCCD 126
Echoes of a Golden Voice	Jasraj	CD	4	1982	IN	Swarashree Enterprises	PJ001
A Life in Music	Amir Khan	CD	2	1984	IN	Saregama	CDNF150641A00
An Undying Parampara Vol. 1	Omikamath Thakur	CD	3	1987	IN	Rhythm House	CD 240361
The Sanctity of Parampara	Manik Bhide	CD	2	1987	IN	Rhythm House	CD 240367
Tribute To Miyan Tansen Vol-1	Vasna Sahasrabudhe	CD	2	1987	IN	Rhythm House	CD 240 369
Tribute To Miyan Tansen Vol-2	Vasna Sahasrabudhe	CD	2	1987	IN	Rhythm House	CD 240 370
An Undying Parampara Vol. 2	Omikamath Thakur	CD	3	1988	IN	Rhythm House	CD 240374
An Undying Parampara Vol. 3	Omikamath Thakur	CD	3	1989	IN	Rhythm House	CD 240381



Musical and advance interfaces

Reactable



Semantic technologies for sound and music

Freesound



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Automatic by relevance ▾

search

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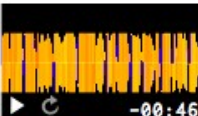
previous next 1 ... 2 3 4 **5** 6 7 8 ... 4667 | 224925 sounds



Sea - Ocean - Beach - Di... ★★★★★
Distant recording of waves crashing on the shore at Boscombe Beach, UK
[field-recording](#) [sandy beach](#) [ocean](#) [sea-shore](#) [waves](#) [sea shore](#) [water](#) [coast](#) [sand](#) [seaside](#)

nebulousflynn
February 25th, 2014
194 downloads
0 comments





Marker Pen on Paper ★★★★★
An assortment of pen sounds for you to chop up and use in your project!
[scribbling](#) [drawing](#) [scribe](#) [marker](#) [signature](#) [paper](#) [pen](#) [write](#) [writing](#) [draw](#)

nebulousflynn
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scots standard english.wav ★★★★★
Me saying 'will you tell the girls about the murder rate of squirrels in third world countries, for a friend ...'
[example voice](#) [english words](#) [scots standard](#) [vocal accent](#) [spoken](#) [fife](#) [male](#)

urbaneguerilla
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tags

1-shot ambience **ambient bass** beat
carnatic compmusic drum
electronic field-recording
h4n-zoom icassp2013-dataset iitm indian-
percussion **loop** metal mridangam
mridangam-stroke-dataset
multisample noise
percussion processed sm-58 snare
sound soundscape **synth** velocity **voice**
water

Semantic technologies for sound and music

AcousticBrainz



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Welcome to MusicBrainz!

MusicBrainz is an open music encyclopedia that collects music metadata and makes it available to the public.

MusicBrainz aims to be:

1. The ultimate source of music information, contributing and releasing the data under a free license.

2. The universal lingua franca for music, providing an unambiguous form of music identification for machines to have meaningful conversations.

Like Wikipedia, MusicBrainz is maintained by a community and we want everyone — including you — to contribute.

[More Information — FAQ](#)

MusicBrainz is operated by the [MetaBrainz Foundation](#), a 501(c)(3) tax-exempt non-profit corporation. MusicBrainz is [free](#) and [open source](#).

Community

 Become a part of our [global community](#) and [start contributing!](#)

Recent Additions

MusicBrainz Blog

Tag Your Music

Latest posts:

- Schema change upgrade instructions, schema 21
- MusicBrainz Picard
- Jaikoz Tagger
- Music MP3 Tagger

AcousticBrainz

Contribute Download Goals API/Data FAQ Detailed Statistics

Welcome to AcousticBrainz!

The AcousticBrainz project aims to crowd source acoustic information for all music in the world and to make it available to the public. This acoustic information describes the acoustic characteristics of music and includes low-level spectral information and information for genres, moods, keys, scales and much more. The goal of AcousticBrainz is to provide music technology researchers and open source hackers with a massive database of information about music. We hope that this database will spur the development of new music technology research and allow music hackers to create new and interesting recommendation engines.

AcousticBrainz is a joint effort between [Music Technology Group](#) at [Universitat Pompeu Fabra](#) in Barcelona and the [MusicBrainz](#) project. AcousticBrainz was originally envisioned by Xavier Serra, the founder and head of the MTG. At the heart of this project lies the [Essentia](#) toolkit from the MTG -- this open source toolkit enables the automatic analysis of music. The output from Essentia is collected by the AcousticBrainz project and made available to the public.

AcousticBrainz organizes the data on a track basis, indexed by the [MusicBrainz ID](#) for [recordings](#). If you know the MBID for a recording, you can easily fetch from AcousticBrainz. For details on how to do this, visit our [API documentation](#).

All of the data contained in AcousticBrainz is licensed under the [CC0 license](#) (public domain).

Examples

If you're wondering what this collected data actually looks like, have a look at the last 5 tracks that have been submitted:

1. [Nickelback / Just For](#)

2. [The National / Bloodbuzz Ohio](#)

3. [Lake of Tears / The Shadowshires](#)

4. [Cornershop / Breaking Every Rule Language English](#)

5. [Limp Bizkit / The Priest](#)

Track Statistics

	Lossy	Lossless	Total
Unique	388862	314427	703289
All	605298	399918	1005216

updated every 10 minutes
last updated: 2014-11-27 17:37
[more on the detailed statistics page](#)

Tweets

 **AcousticBrainz**
@AcousticBrainz 2h

We have so many files our counter has overflowed its box! Thanks to everyone for 1m submissions and 700k uniques
[pic.twitter.com/SmNHfEock](#)

all music in the world is described by the acoustic information for each track. This is to provide music technology researchers and open source hackers with a massive database of information about music. The output from Essentia is collected by the AcousticBrainz project and made available to the public.

envisaged by Xavier Serra

Track Statistics

	Lossy	Lossless	Total
Unique	388519	312628	701147
All	603004	396717	1000321

updated every 10 minutes

References and credits

- <http://upf.edu>, <http://www.upf.edu/dtic/>, <http://mtg.upf.edu>
- <http://en.wikipedia.org/wiki/Vocaloid>
- <http://compmusic.upf.edu/>
- <http://en.wikipedia.org/wiki/Reactable>
- <http://acousticbrainz.org/>
- Slides released under CC Attribution-Noncommercial-Share Alike license and available from <https://github.com/MTG/sms-tools>

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